6/04/26

# Quiz-5

Find the Fourier Transform of the following signals:

1. (7 marks)

$$x(t) = t \cdot e^{-t} \cos(t) \cdot u(t).$$

2. (3 marks)

$$x(t) = \exp[t^4].$$

I. LTI systems:

(i) Stable

(ii) Input-output relationship is characterized by a linear constant coefficient differential equation.

$$\sum_{k=0}^{N} a_k \cdot y^{(k)}(t) = \sum_{l=0}^{M} b_l \cdot x^{(l)}(t)$$

Ex: $y'' + 4y' + 3y = x' + 2x$ (from Chap. 4 of OWN).

Apply Fourier transform on both sides:

$$\sum_{k=0}^{N} a_k (j\omega)^k Y(\omega) = \sum_{l=0}^{M} b_l (j\omega)^l X(\omega).$$

$$Y(\omega) = X(\omega) \cdot \left[ \frac{\sum_{l=0}^{M} b_l (j\omega)^l}{\sum_{k=0}^{N} a_k \cdot (j\omega)^k} \right]$$

$$Y(\omega) = X(\omega) \cdot H(\omega).$$

$$H(\omega) = \frac{\sum_{l=0}^{M} b_l \cdot (j\omega)^l}{\sum_{k=0}^{N} a_k \cdot (j\omega)^k}.$$

Example: $y'' + 4y' + 3y = x' + 2x.$

Assuming that the system is stable and LTI:

$$(j\omega)^2 Y + 4j\omega \cdot Y + 3Y = j\omega \cdot X + 2X$$

$$\Longrightarrow \quad H = \frac{Y}{X} = \frac{j\omega + 2}{(j\omega)^2 + 4j\omega + 3}$$

$$= \frac{j\omega + 2}{(j\omega + 1)(j\omega + 3)}$$

$$= \frac{A}{j\omega + 1} + \frac{B}{j\omega + 3}$$

It is easy to see that $A = B = 1/2$

$$\Rightarrow \quad H(\omega) = \frac{1/2}{j\omega + 1} + \frac{1/2}{j\omega + 3}$$

$$\Rightarrow \quad h(t) = \frac{1}{2} \cdot e^{-t} u(t) + \frac{1}{2} \cdot e^{-3t} \cdot u(t).$$

## II. Fourier Transforms of Distributions.

<u>Remark:</u> If $x(t)$ is Schwartz then:

(i) $X(\omega)$ is Schwartz,

(ii) Fourier inversion theorem applies to $x \overset{F}{\leftrightarrow} X$.

$$\frac{1}{2\pi}\int X(\omega)e^{j\omega t}d\omega = x(t) \quad \forall t.$$

<u>Remark:</u> Why do we need distributions and their Fourier transf?

Some LTI systems have a distribution as their impulse response, such as:

(i) $x \to \boxed{\phantom{x}} \to x \quad : \quad h(t) = \delta(t)$

(ii) $x \to \boxed{\phantom{x}} \to x'(t) : \quad h(t) = \delta'(t)$

(iii) $x \to \boxed{\phantom{x}} \to y(t) = \int_{-\infty}^{t} x(\tau)d\tau : \quad h(t) = u(t).$

It can be easier to work with these LTI systems in the frequency domain via Fourier transforms.

For instance, example (ii) in frequency domain is:

$$Y(\omega) = j\omega \cdot X(\omega).$$